DEVELOPMENT OF FACE

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Refereces:
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Learning objectives

- Concerning the development of the face, the student shall be able to:
- 1) describe the mesenchymal facial processes around the developing mouth (stomodeum)
- 2) understand how these facial processes contribute to the formation of the upper and lower lip regions
- 3) Describe the ectodermal placodes on the developing face and know their derivatives
- 4) understand how disturbances in normal development can result in common congenital abnormalities (e.g., clefts of the lip)

4 th week I.U

- During the 4th week of development in utero, the primitive oral cavity (stomodeum) is bounded by five facial swellings that are produced by proliferating zones of mesenchyme (neural crest tissue) lying beneath the surface ectoderm – the frontonasal, mandibular and maxillary processes. The frontonasal process lies above, the two mandibular processes lie below, and the two maxillary processes are located at the sides.
- The maxillary and mandibular processes are derived from the first pharyngeal arches. A membrane called the oropharyngeal membrane separates the primitive oral cavity from the developing pharynx

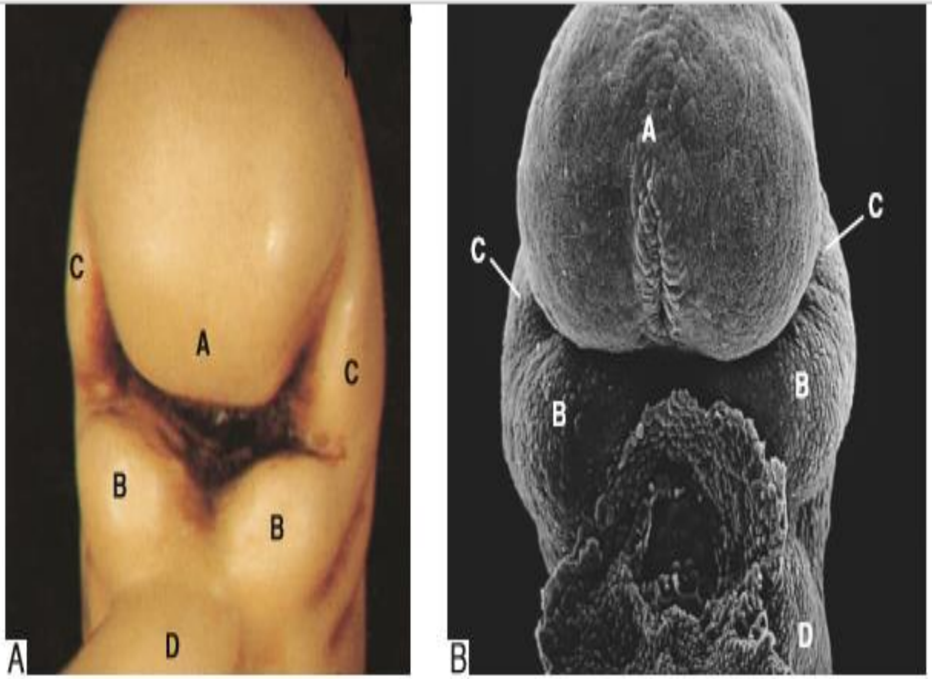


Fig. 17.1 (A) Model of a face of a 4-week-old human embryo (frontal aspect). (B) SEM of the face at an equivalent stage to 4 weeks in the rat (frontal aspect). A = frontonasal process; B = mandibular processes; C = maxillary processes; D = pericardial swelling. Courtesy Prof. A G S Lumsden.

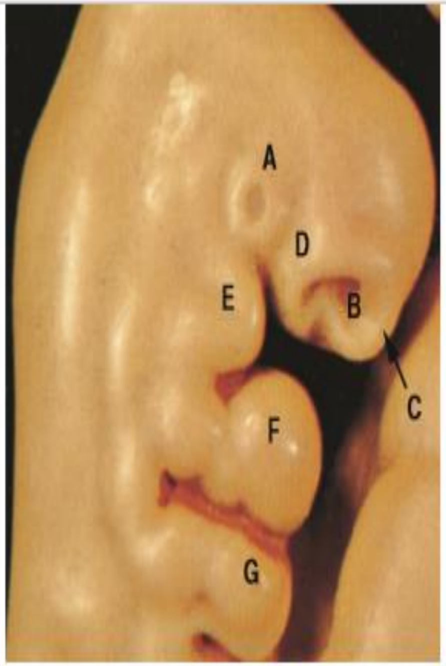


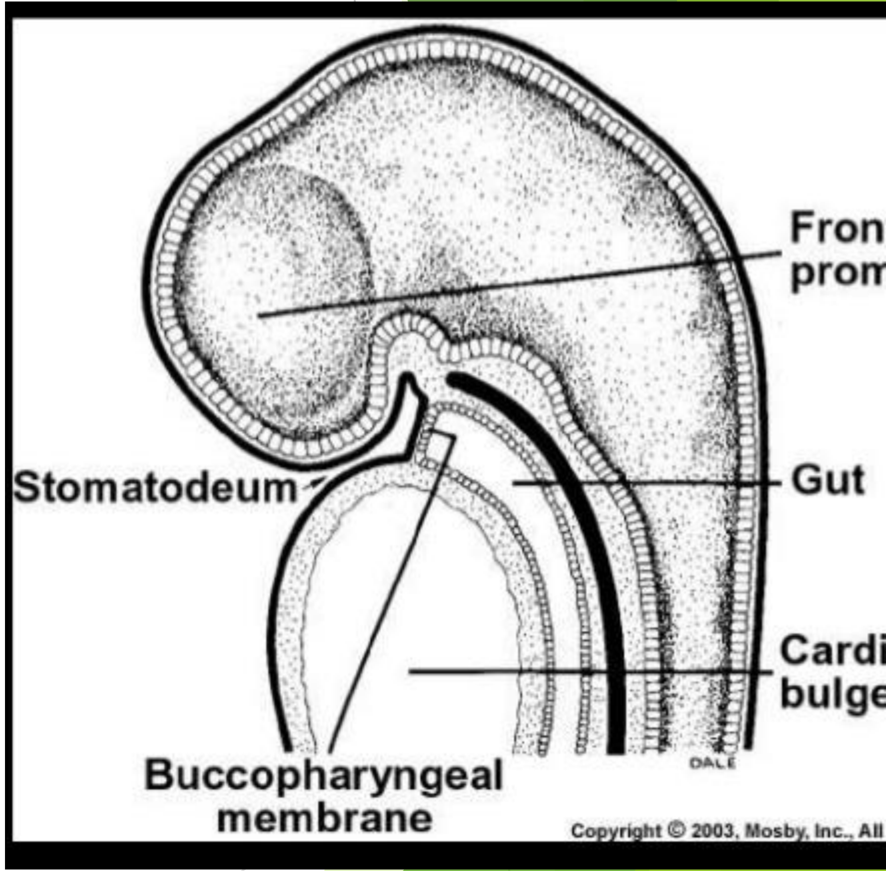
Fig. 17.2 Model of a face of a 5-week-old human embryo (lateral aspect). A = optic placode; B = nasal pit; C = medial nasal process; D = lateral nasal process; E = maxillary process; F = mandibular process; G = second pharyngeal arch.

INTRODUCTION

- The face develops in the human between the 4th and 10th weeks of intra-uterine life.
- Facial development is a complex, multistep process, involving facial processes containing neural crest mesenchyme and sensory facial ectodermal placodes.
- There are epithelial–mesenchymal interactions and coordination from the sequential activation of specific genes and signalling molecules

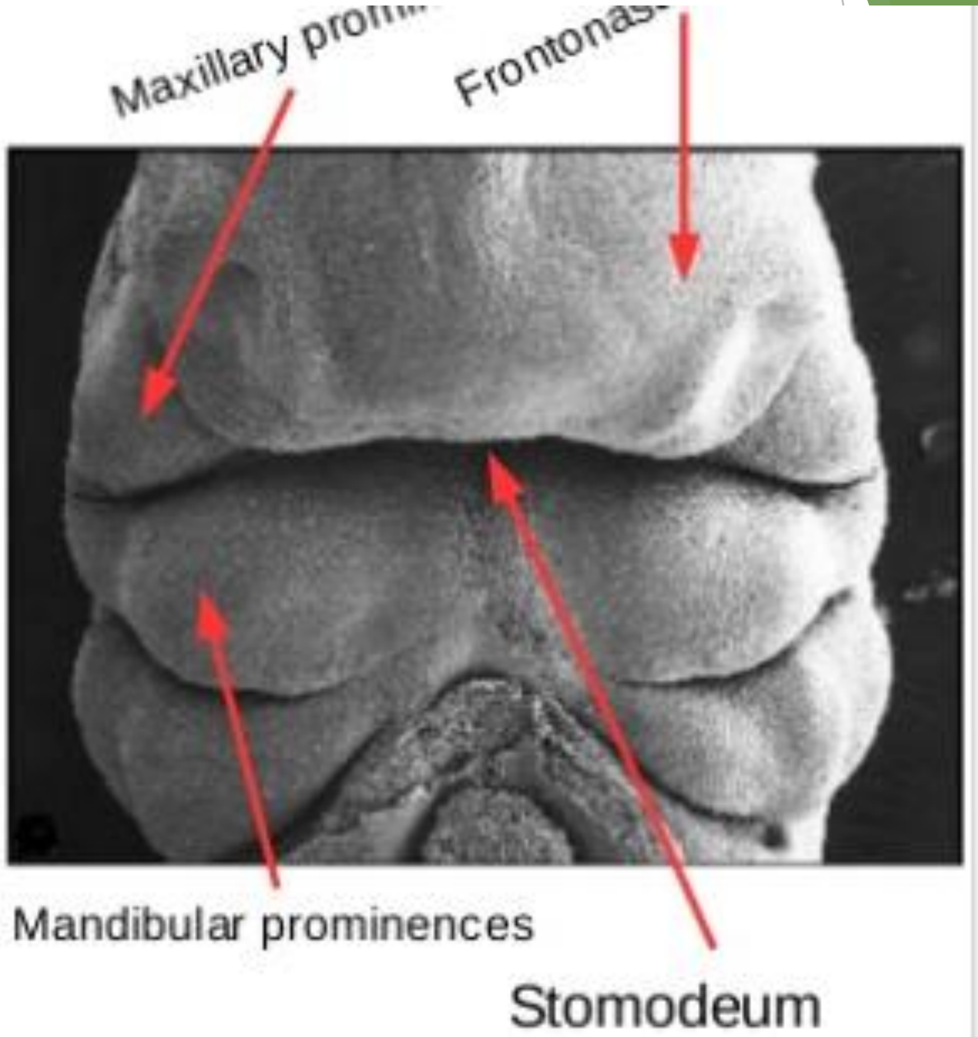
OROPHARYNGEAL MEMBRANE

- This membrane is bilaminar, being composed of an outer ectodermal layer and an inner endodermal layer.
- The oropharyngeal membrane soon breaks down to establish continuity between the ectoderm-lined oral cavity and the endoderm-lined pharynx.
- Although not detectable in the adult, the demarcation zone between mucosa derived from ectoderm and endoderm is said to correspond to a region lying just behind the permanent third molar tooth.



4^{th} week intra uterine

- When the embryo is $4^{1/2}$ weeks old five mesenchymal prominenses can be recognised.
- The frontonasal prominence, a slightly rounded elevation cranial to the stomodeum.
- The maxillary prominences (dorsal portion of the 1^{st} arch), lateral to the stomodeum.
- The mandibular prominences, caudal to the stomodeum.
- Development of face is later complemented by the formation of the nasal prominences.



5th week I.U.

- In the 5th week, ectodermal thickenings give rise to the nasal and optic placodes, forming the olfactory epithelium and the lenses of the eyes respectively.
- The nasal placodes sink into the underlying mesenchyme, creating two blind-ended nasal pits.
- Proliferation of mesenchyme from the frontonasal process around the openings of the nasal pits produces the medial and lateral nasal processes.
- In addition, the maxillary processes enlarge and grow forwards and medially.

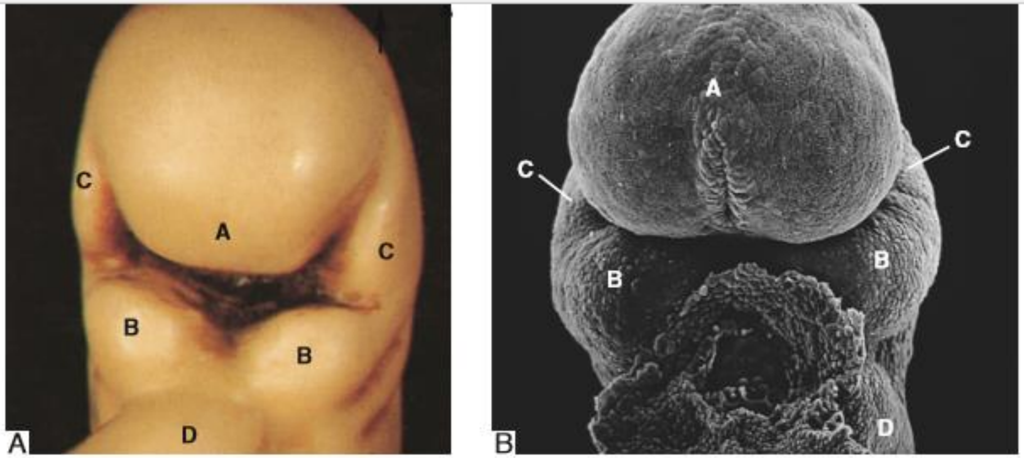


Fig. 17.1 (A) Model of a face of a 4-week-old human embryo (frontal aspect). (B) SEM of the face at an equivalent stage to 4 weeks in the rat (frontal aspect). A = frontonasal process; B = mandibular processes; C = maxillary processes; D = pericardial swelling. Courtesy Prof. A G S Lumsden.

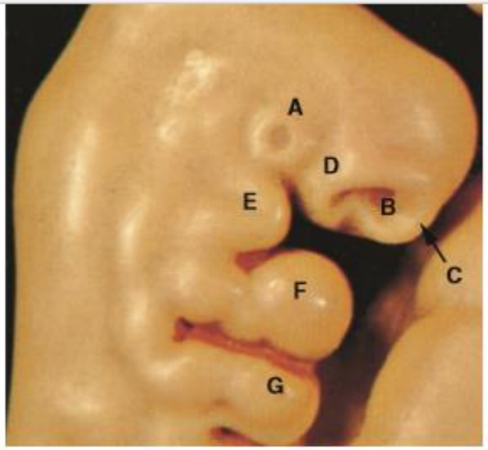


Fig. 17.2 Model of a face of a 5-week-old human embryo (lateral aspect). A = optic placode; B = nasal pit; C = medial nasal process; D = lateral nasal process; E = maxillary process; F = mandibular process; G = second pharyngeal arch.

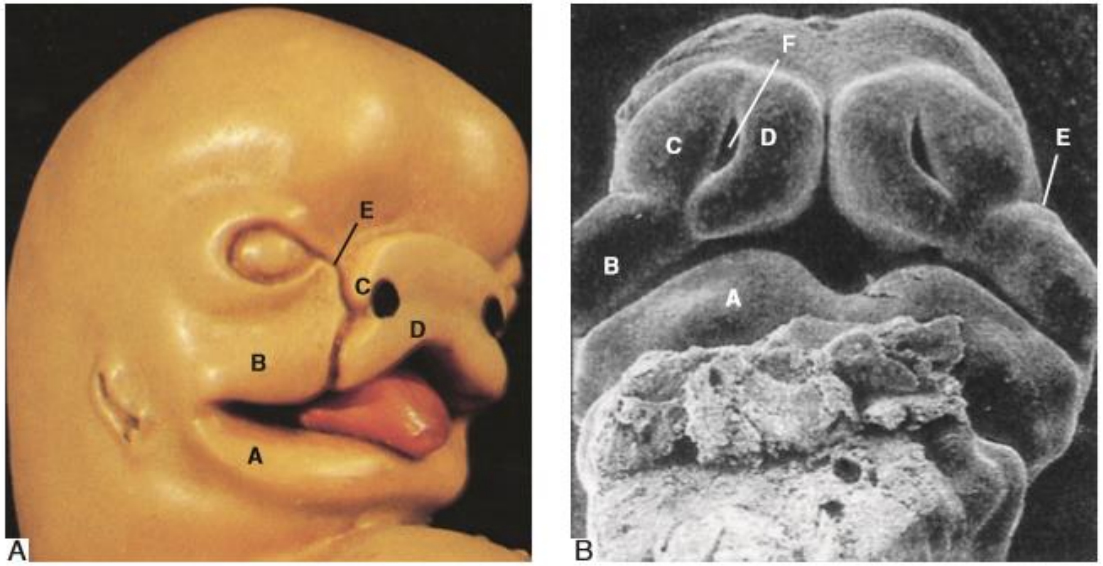


Fig. 17.3 (A) Model of a face of a 6-week-old human embryo (lateral aspect). A = mandibular process; B = maxillary process; C = lateral nasal process; D = medial nasal process; E = naso-optic furrow. (B) SEM of developing rat upper jaw and lip at an equivalent stage to 6 weeks showing the merging maxillary and medial nasal processes. Labelling as for Fig. 17.3A; F = nasal pit. Courtesy Prof. A G S Lumsden.

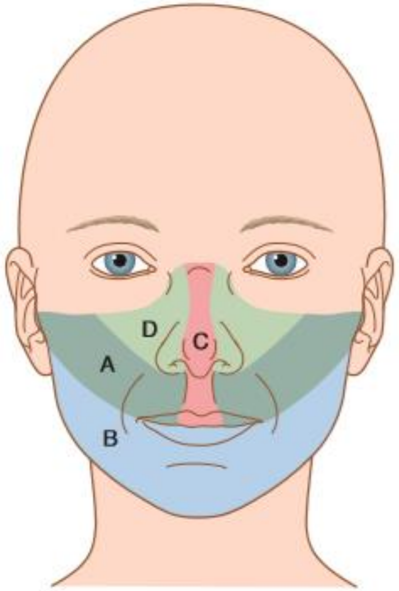
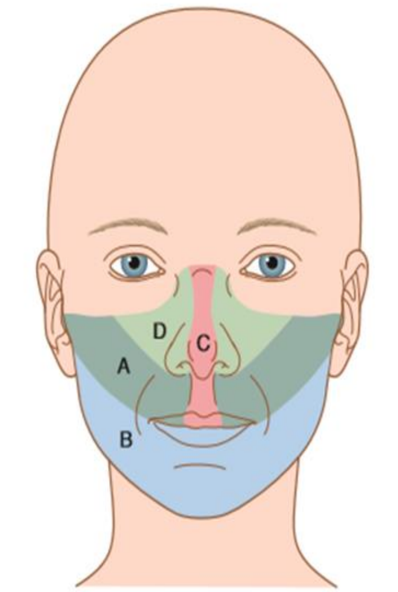


Fig. 17.4 The contributions to the adult face from the embryonic facial processes. A = maxillary processes; B = mandibular processes; C = medial nasal processes; D = lateral nasal processes.

6^{th} week.I.U

- By the 6th week,
- the two mandibular processes fuse in the midline to form the tissues of the lower jaw.
- The mandibular processes and maxillary processes meet at the angles of the mouth.
- From the corners of the mouth, the maxillary processes grow inwards beneath the lateral nasal processes and towards the medial nasal processes of the upper lip.
- Between the merging maxillary and the lateral nasal processes lie the naso-optic furrows that will eventually form the nasolacrimal ducts.
- The maxillary processes subsequently overgrow the medial nasal processes to meet in the midline and thus contribute all the tissue for the upper lip.



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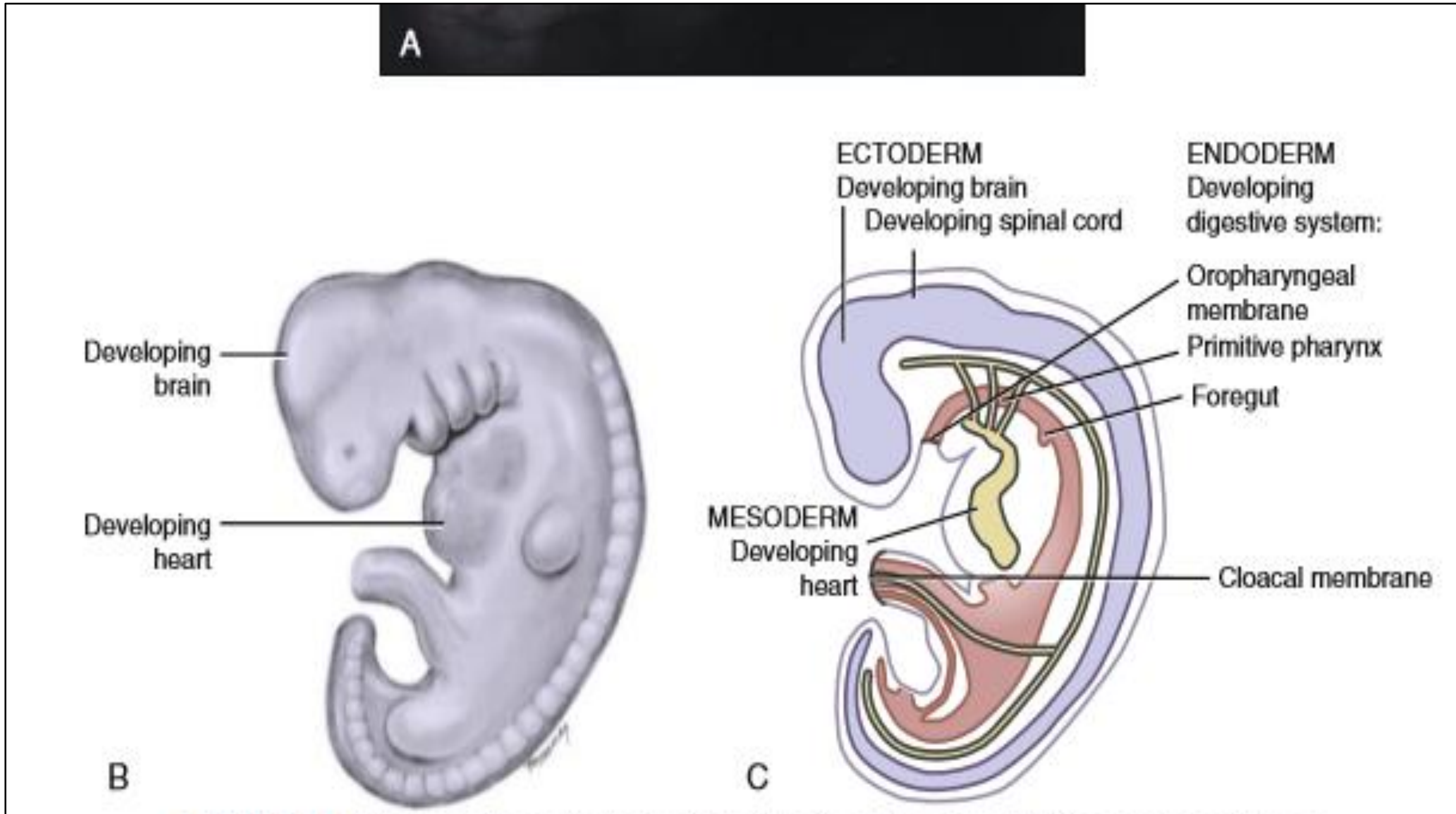


FIGURE 3-14 Trilaminar embryonic disc has folded into the embryo. **A** and **B,** Disc now has undergone embryonic folding as a result of extensive growth of the ectoderm, and there is development of the brain with spinal cord, heart, and digestive tract. **C,** With this folding, the endoderm is now inside the ectoderm, with the mesoderm filling in the areas between the two tissue types, except at the two embryonic membranes as shown on cross section. (From Moore KL, Persaud TVN, Torchia MG: *The developing human: clinically oriented embryology*, ed 10, St Louis, 2015, Saunders/Elsevier.)

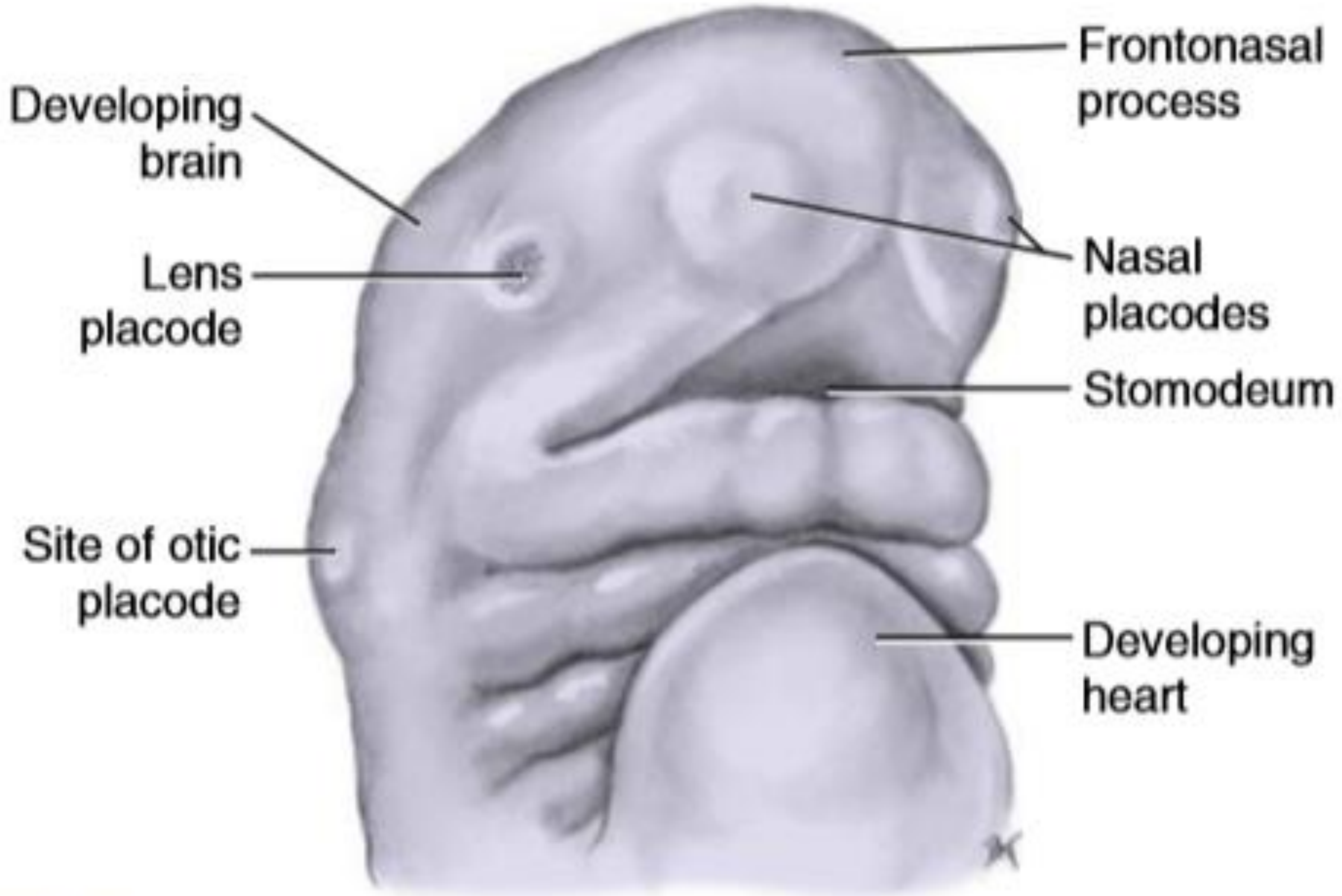


FIGURE 4-2 Embryo at the fourth week of prenatal development showing the developing brain, the forming face from the growth of

Developmental Events Overview

- Disintegration of the oropharyngeal membrane of stomodeum enlarges the primitive mouth, allowing access to the primitive pharynx
- • Mandibular processes fuse to form mandibular arch, which then uses to form the mandible and lower lip
- • Frontonasal process forms and gives rise to the nasal placodes, nasal pits, medial and lateral nasal processes, and intermaxillary segment to form the nose and primary palate
- • Maxillary process forms from mandibular arch on each side of the stomodeum
- • Maxillary processes fuse with each medial nasal process to form upper lip and with each mandibular arch to form the labial commissures

EMBRYONIC STRUCTURES	ORIGIN	FUTURE STRUCTURES
Stomodeum	Ectodermal depression enlarged by disintegration of oropharyngeal membrane	Oral cavity proper
First branchial arch (aka *mandibular arch*)	Fused mandibular processes and neural crest cells	Lower lip, lower face, mandible with associated tissue (other arch derivatives shown in Table 4-2)
Maxillary process(es)	Superior and anterior swelling(s) from mandibular arch and neural crest cells	Midface, upper lip sides, cheeks, secondary palate, posterior part of maxilla with associated tissue, zygomatic bones, part of temporal bones
Frontonasal process	Ectodermal tissue and neural crest cells	Medial and lateral nasal processes
Nasal pits	Nasal placodes	Nasal cavities
Medial nasal process(es)	Frontonasal process medial to nasal pits	Middle of nose, philtrum region, intermaxillary segment
Intermaxillary segment	Fused medial nasal processes	Anterior part of maxilla with associated tissue, primary palate, nasal septum
Lateral nasal process(es)	Frontonasal process lateral to nasal pits	Nasal alae

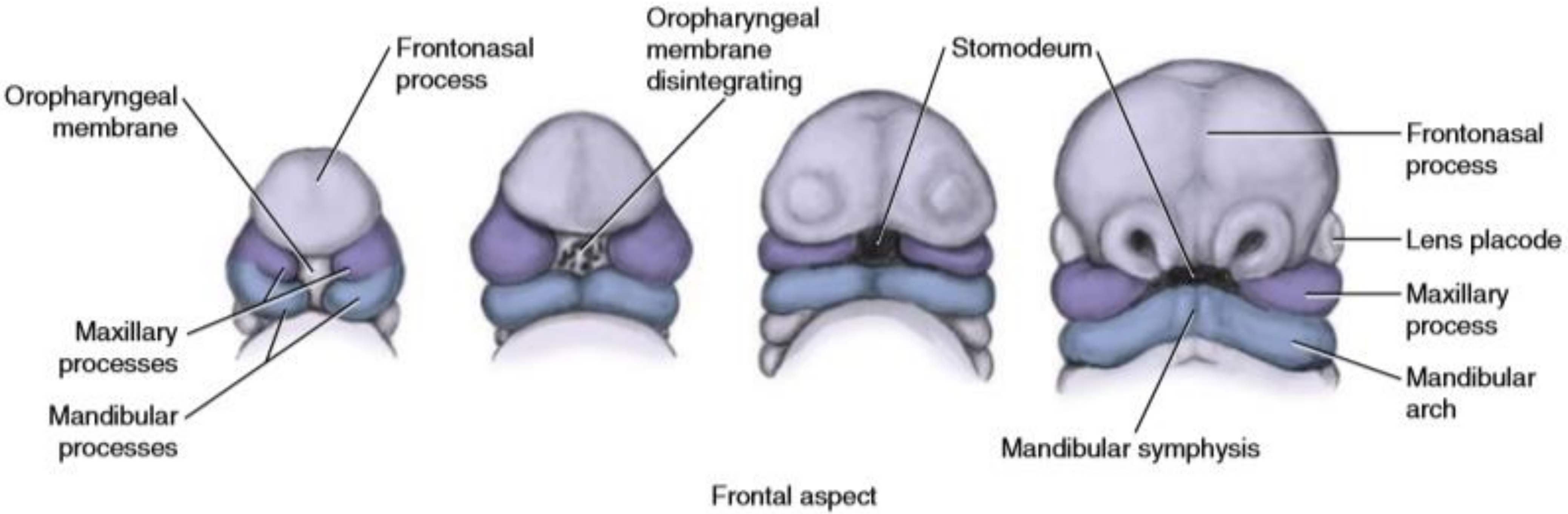


FIGURE 4-5 During the third to fourth week, disintegration of the oropharyngeal membrane enlarges the stomodeum of the embryo and allows access between the primitive mouth and the primitive pharynx. The frontonasal process also enlarges, helping to form the nasal region. Mandibular processes give rise to the maxillary processes and then fuse together at the mandibular symphysis, forming the mandibular arch inferior to the enlarged stomodeum.

Prominence	Structures Formed
Frontonasal[a]	Forehead, bridge of nose, medial and lateral nasal prominences
Maxillary	Cheeks, lateral portion of upper lip
Medial nasal	Philtrum of upper lip, crest and tip of nose
Lateral nasal	Alae of nose
Mandibular	Lower lip

[a] The frontonasal prominence is a single unpaired structure; the other prominences are paired.

Intermaxillary segment

- Two differing accounts have been given for the continued development of the upper lip beyond the 6th week of intrauterine life.
- One view suggests that the maxillary processes overgrow the medial nasal processes

Nasolacrimal duct

- • Develops from a rod-like thickening of the ectoderm in the floor of the nasolacrimal groove •
- This solid cord of cells separates from the surface ectoderm and lies in the underlying mesenchyme

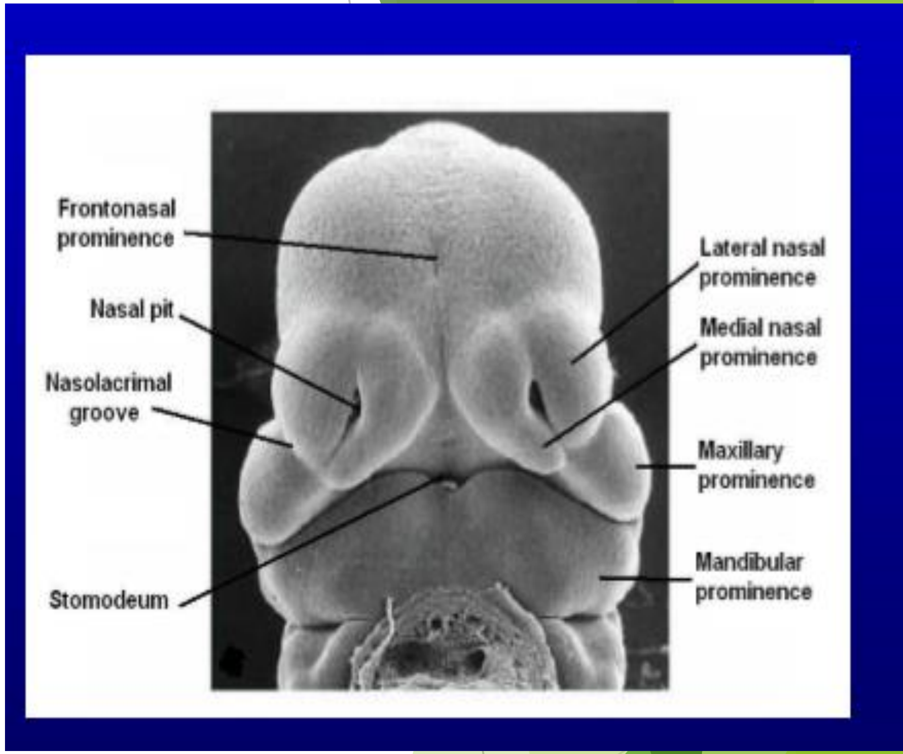
• The cord gets canalized to form the nasolacrimal duct • The cranial end of the duct expands to form the lacrimal sac •

The caudal end opens into the inferior meatus of the nasal cavity •

The duct is usually becomes completely patent only after birth

• Failure of complete canalization of the duct leads to atresia of the duct (seen in about 6% of newborn infants)

- With the formation of the medial and lateral nasal prominences, the nasal placodes lie in the floor of depressions called the nasal pits
- • By the end of 6th week, nasal pits deepen and form nasal sacs With the formation of the medial and lateral nasal prominences, the nasal placodes lie in the floor of depressions called the nasal pits
- • By the end of 6th week, nasal pits deepen and form nasal sacs
- • Each nasal sac grows dorsocaudally, ventral to the developing brain



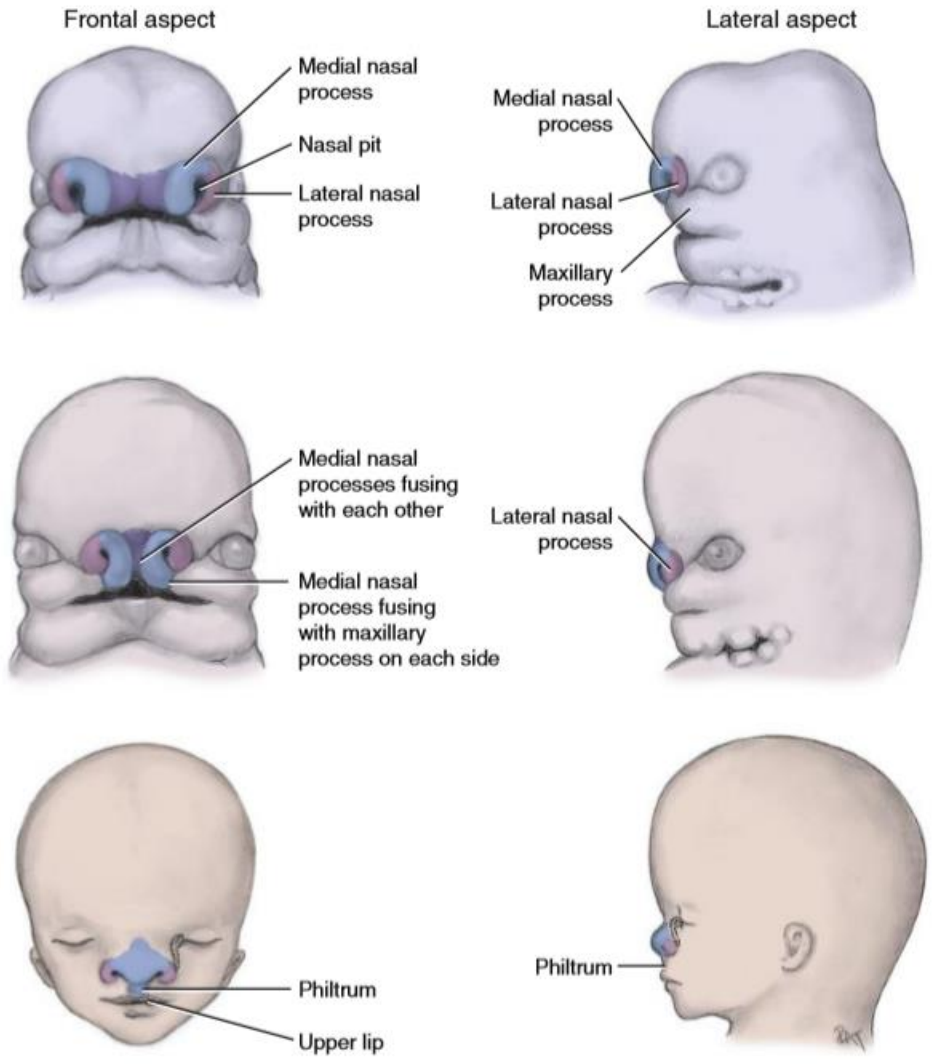


FIGURE 4-6 Development of the nose from the fusion of the medial and lateral nasal processes; also showing the formation of the upper lip from the medial process fusing with the maxillary process on each side of the stomodeum.

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TERATOGENS

- hromosomal abnormality (syndromic hydroprosencephaly) or a gene mutation (nonsyndromic hydroprosencephaly).
- Among environmental factors that may be relevant are maternal diabetes, infections during pregnancy (e.g., syphilis, rubella)
- drugs during pregnancy (e.g., retinoic acid, alcohol, anticonvulsants

MICROSTOMIA

1Holoprosencephaly

- This is a congenital abnormality in which the developing forebrain fails to divide into two separate hemispheres and ventricles

2

Disturbances in the relationship between the mandibular and maxillary processes may give rise to

- macrostomia (enlarged oral orifice),
- microstomia (small oral orifice) or,
- rarely, astomia (lack of an oral orifice).

MACROSTOMIA

Treacher Collins' syndrome

- Treacher Collins' syndrome is found in 1 : 10,000 live births
- . Typically, the syndrome is characterised by downward-slanting eyes, coloboma of the eye (a notchlike defect with a keyhole appearance in the inferior part of the iris resulting from failure of closure of the retinal fissure of the developing eye), micrognathia, absent or malformed ears and clefts of the lips and palate.

TREACHER COLLINS SYNDROME

FIGURE 3-16 Treacher Collins syndrome from the failure of migration of the neural crest cells to the facial region in the embryo presents with areas without complete orofacial development, having marked features, including micrognathia (small lower jaw). (From Kaban LB, Toulis MJ: *Pediatric oral and maxillofacial surgery*, ed 1, Philadelphia, 2004, Saunders/Elsevier.)

Fetal alcoholic syndrome

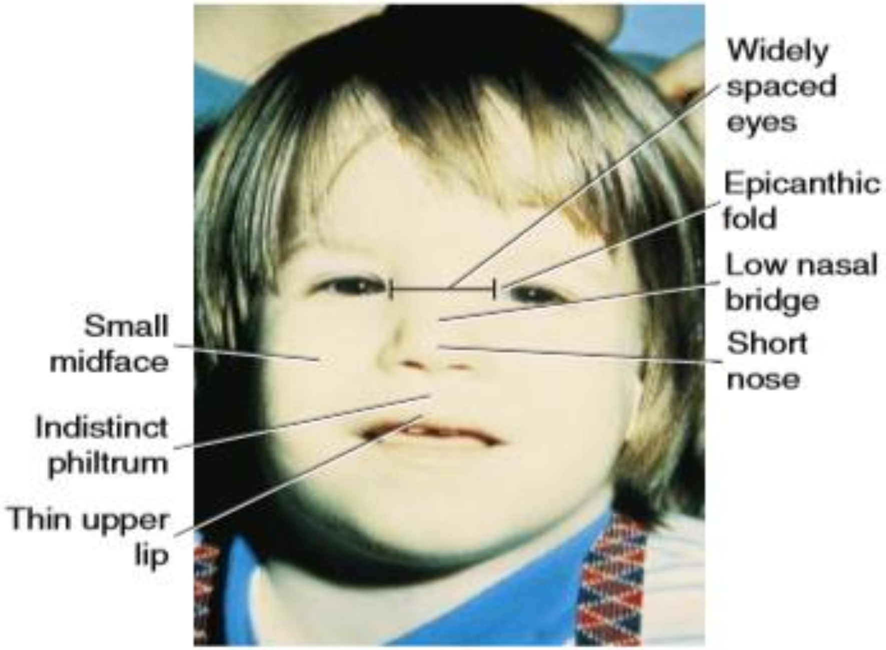
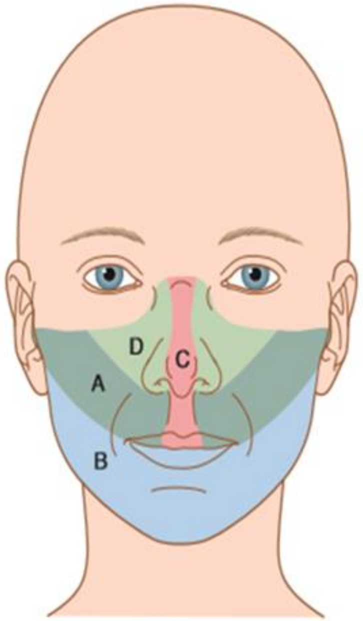


FIGURE 3-18 Fetal alcohol syndrome presents with noted orofacial features and various levels of mental disability. This syndrome is caused by the pregnant women's excessive use of ethanol during the embryonic period. (From Streissguth AP, Landesman-Dwyer S, Martin JC, et al: Teratogenic effects of alcohol in humans and laboratory animals, *Science* 209:353-361, 1980. Copyright 1980; American Association for the Advancement of Science.)

Intermaxillary segment.



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CLEFT LIP

- The mesoderm of the lateral part of the lip is formed from the maxillary process. The overlying skin is derived from ectoderm of the same process.
- The failure of fusion of medial nasal process with the lateral nasal process leads to the formation of cleft lip.

Video links for extended learning

- https://youtu.be/DgZ_tqucdI4